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/**
 * 🧩 Inventory Manager with Generics — Student Template
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 * Task Description (2–3 lines):
 * Build a small Kotlin console program that manages an inventory of items (like a store or warehouse).
 * Use lists, maps, and generic functions to handle data and print filtered results.
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 * Instructions:
 * 1) Create a data class `Item` with properties: name (String), category (String), and quantity (Int).
 * 2) Create a function `printInventory(items: List<Item>)` to print all items in a clean table-like format.
 * 3) Create a **generic function** `filterList()` that filters any list using a provided condition (lambda).
 *   - Example signature: `fun <T> filterList(list: List<T>, condition: (T) -> Boolean): List<T>`
 * 4) In `main()`:
 *   - Create a list of sample items (5–6) across multiple categories.
 *   - Use `filterList()` to get:
 *       a) All items with quantity < 10 (low stock)
 *       b) All items in a specific category (e.g., "Electronics")
 *   - Print results using `printInventory()`.
 * 5) BONUS (Optional): Create a map showing total quantities per category using `groupBy` and `sumOf`.
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 * 💡 Hints:
 * - Use string templates and formatting for clean output.
 * - Practice using higher-order functions like `filter`, `map`, and `groupBy`.
 * - Generics help make your filter function reusable across different data types.
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*  Sample Output:
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* --- Full Inventory ---
* Name: Laptop | Category: Electronics | Qty: 12
* Name: Keyboard | Category: Electronics | Qty: 5
* Name: Chair | Category: Furniture | Qty: 20
* Name: Table | Category: Furniture | Qty: 8
* Name: Notebook | Category: Stationery | Qty: 50
*
* --- Low Stock (<10) ---
* Name: Keyboard | Category: Electronics | Qty: 5
* Name: Table | Category: Furniture | Qty: 8
*
* --- Electronics Category ---
* Name: Laptop | Category: Electronics | Qty: 12
* Name: Keyboard | Category: Electronics | Qty: 5
*
* --- Total Quantity per Category ---
* Electronics: 17
* Furniture: 28
* Stationery: 50
*
* TODOs:
* // TODO: Create data class Item(name, category, quantity)
* // TODO: Implement printInventory(items)
* // TODO: Implement generic function filterList()
* // TODO: In main(), create list and demonstrate filtering and grouping
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// ----------------------
// Student Code Area
// ----------------------

// TODO: Define data class Item(name: String, category: String, quantity: Int)

// TODO: Implement printInventory(items: List<Item>)

// TODO: Implement generic filterList(list: List<T>, condition: (T) -> Boolean): List<T>

// TODO: Implement main() — create sample items, call filterList() & print results
```

```
data class Item(
    val name: String,
    val category: String,
    val quantity: Int
)

fun printInventory(items: List<Item>) {
    if (items.isEmpty()) {
        println("No items found.")
        return
    }

    for (item in items) {
        println("Name: ${item.name} | Category: ${item.category} | Qty: ${item.quantity}")
    }
}

fun <T> filterList(list: List<T>, condition: (T) -> Boolean): List<T> {
    val result = mutableListOf<T>()

    for (element in list) {
        if (condition(element)) {
            result.add(element)
        }
    }

    return result
}
```

```
fun main() {

    val inventory = listOf(
        Item("Laptop", "Electronics", 12),
        Item("Keyboard", "Electronics", 5),
        Item("Chair", "Furniture", 20),
        Item("Table", "Furniture", 8),
        Item("Notebook", "Stationery", 50),
        Item("Pen", "Stationery", 6)
    )

    println("\n--- Full Inventory ---")
    printInventory(inventory)

    val lowStock = filterList(inventory) { it.quantity < 10 }

    println("\n--- Low Stock (<10) ---")
    printInventory(lowStock)

    val electronicsItems = filterList(inventory) { it.category == "Electronics" }

    println("\n--- Electronics Category ---")
    printInventory(electronicsItems)

    val totalPerCategory = inventory
        .groupBy { it.category }
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        .mapValues { entry -> entry.value.sumBy { it.quantity } }

    println("\n--- Total Quantity per Category ---")
    for ((category, totalQty) in totalPerCategory) {
        println("$category: $totalQty")
    }
}
```